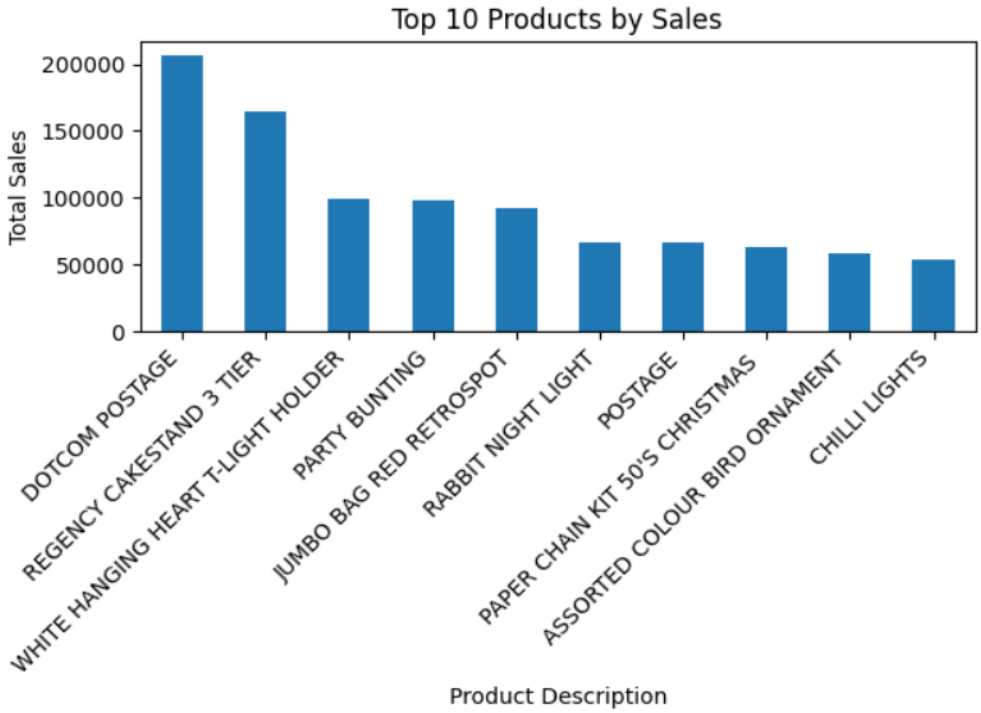
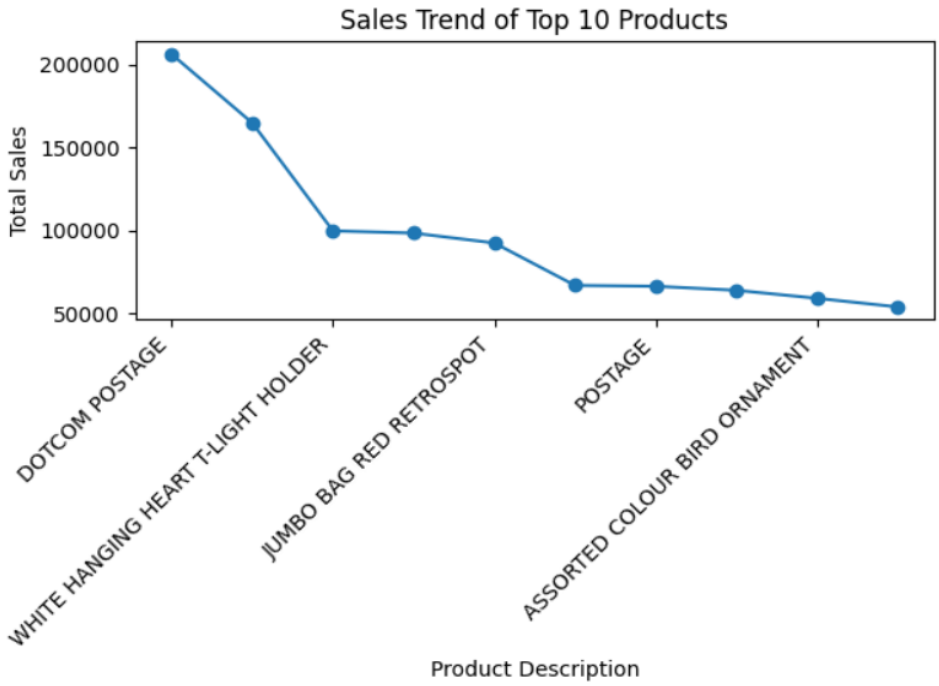
Introduction to Python Libraries and Dataset Preprocessing

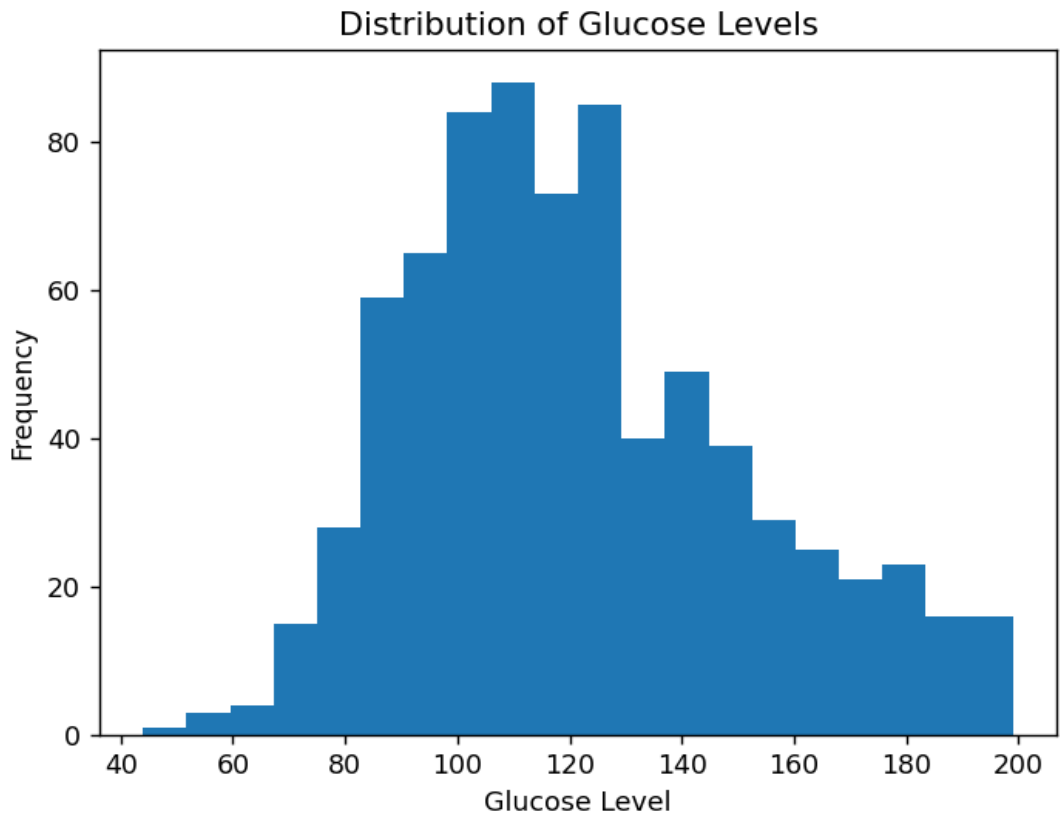
SCENARIO 1: E-Commerce Sales Data

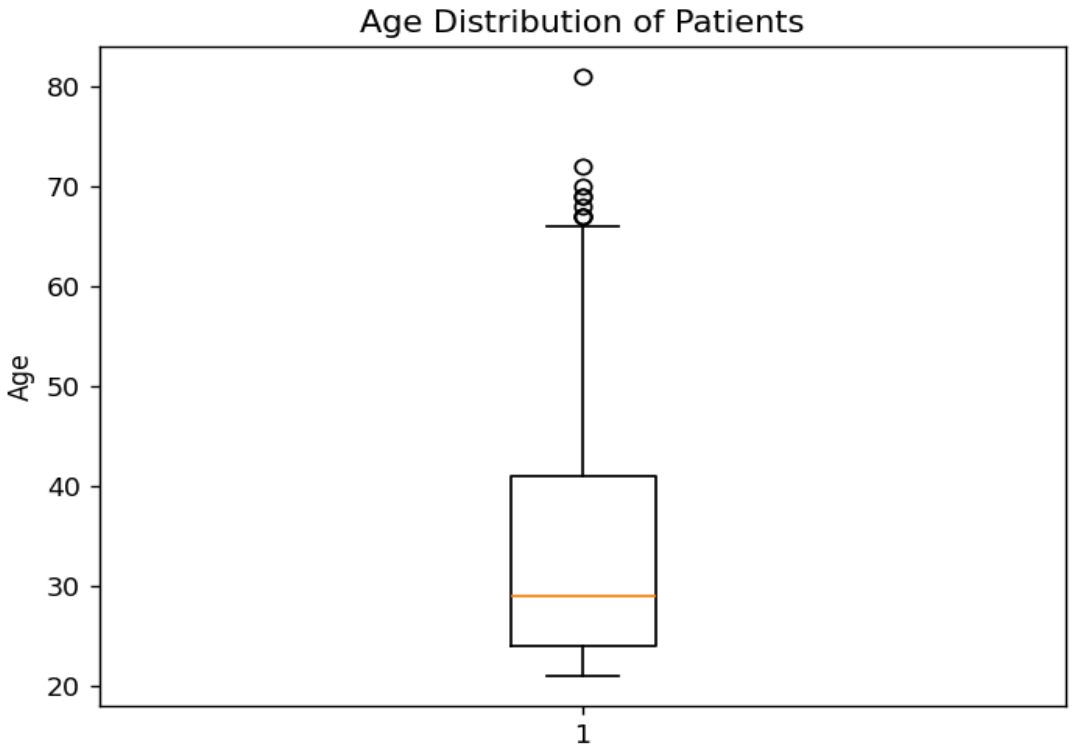
Scenario – 1:



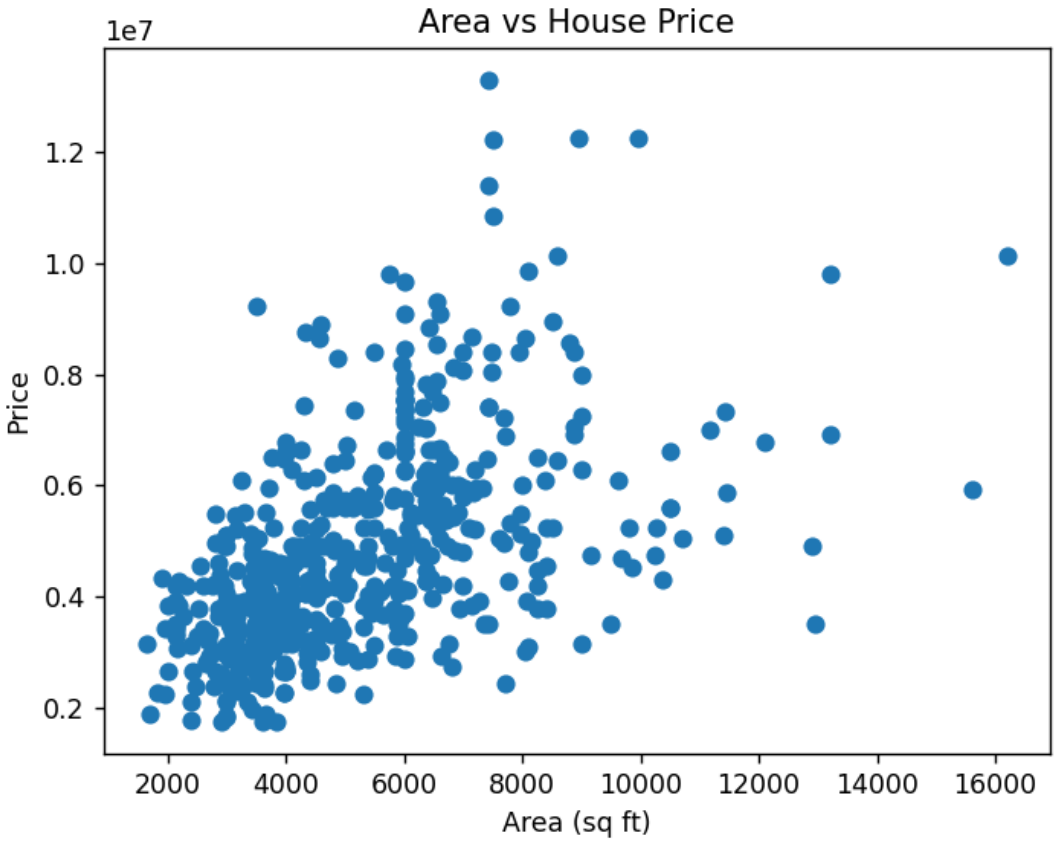


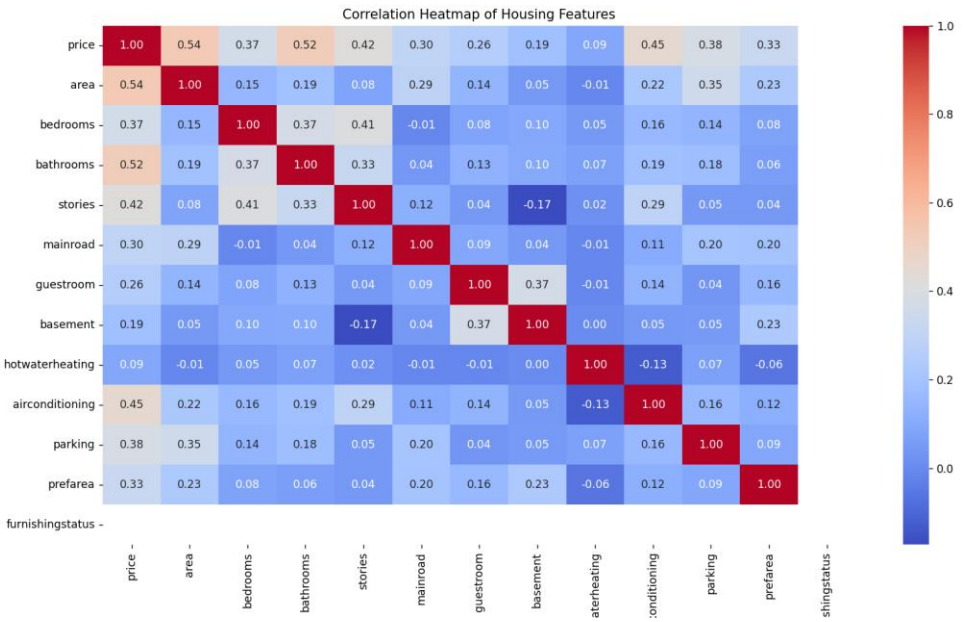
Scenario – 2:


Distribution of Glucose Levels
Frequency
0
20
40
60
80
40
60
80
100
120
140
160
180
200
Glucose Level

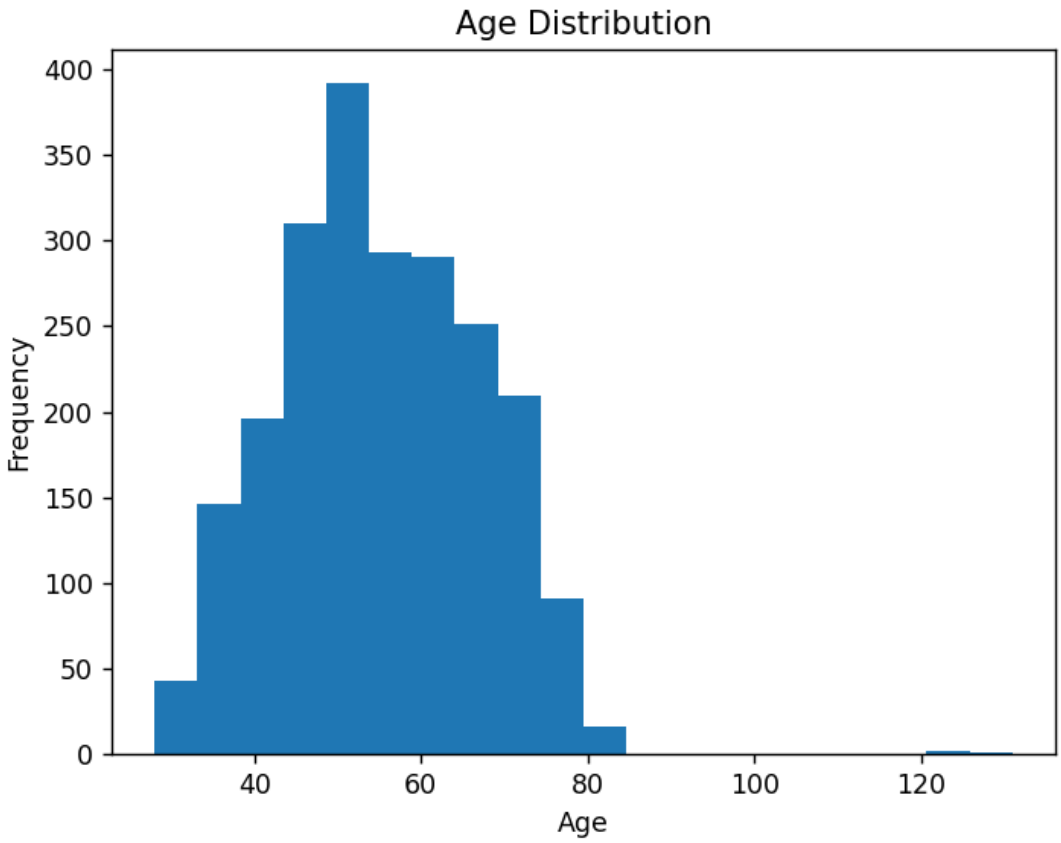

Age Distribution of Patients
Age
20
30
40
50
60
70
80
1

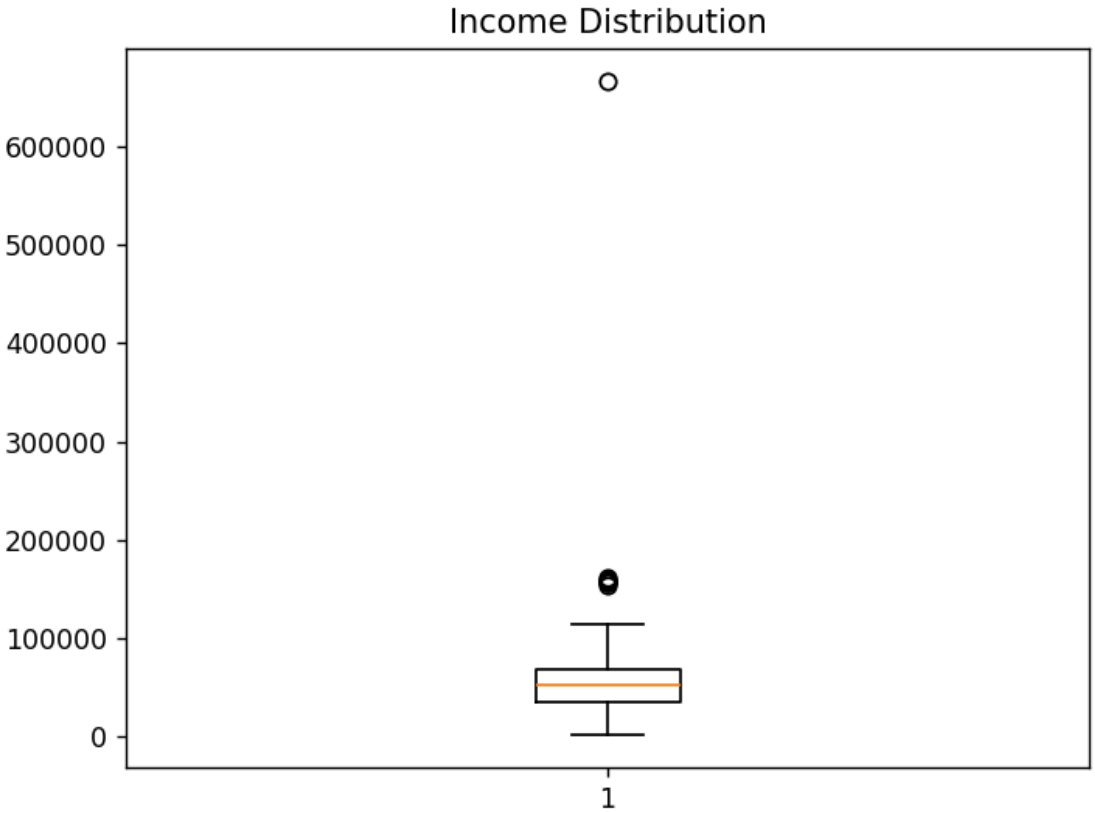
Scenario – 3:


Area vs House Price
1e7
1.2
1.0
0.8
0.6
0.4
0.2
Price
2000
4000
6000
8000
10000
12000
14000
16000
Area (sq ft)


Correlation Heatmap of Housing Features
price 1.00 0.54 0.37 0.52 0.42 0.30 0.26 0.19 0.09 0.45 0.38 0.33
area 0.54 1.00 0.15 0.19 0.08 0.29 0.14 0.05 -0.01 0.22 0.35 0.23
bedrooms 0.37 0.15 1.00 0.37 0.41 -0.01 0.08 0.10 0.05 0.16 0.14 0.08
bathrooms 0.52 0.19 0.37 1.00 0.33 0.04 0.13 0.10 0.07 0.19 0.18 0.06
stories 0.42 0.08 0.41 0.33 1.00 0.12 0.04 -0.17 0.02 0.29 0.05 0.04
mainroad 0.30 0.29 -0.01 0.04 0.12 1.00 0.09 0.04 -0.01 0.11 0.20 0.20
guestroom 0.26 0.14 0.08 0.13 0.04 0.09 1.00 0.37 -0.01 0.14 0.04 0.16
basement 0.19 0.05 0.10 0.10 -0.17 0.04 0.37 1.00 0.00 0.05 0.05 0.23
hotwaterheating 0.09 -0.01 0.05 0.07 0.02 -0.01 -0.01 0.00 1.00 -0.13 0.07 -0.06
airconditioning 0.45 0.22 0.16 0.19 0.29 0.11 0.14 0.05 -0.13 1.00 0.16 0.12
parking 0.38 0.35 0.14 0.18 0.05 0.20 0.04 0.05 0.07 0.16 1.00 0.09
prefarea 0.33 0.23 0.08 0.06 0.04 0.20 0.16 0.23 -0.06 0.12 0.09 1.00
furnishingstatus
price
area
bedrooms
bathrooms
stories
mainroad
guestroom
basement
aterheating
onditioning
parking
prefarea
shingstatus
1.0
0.8
0.6
0.4
0.2
0.0

Scenario – 4:


Age Distribution
Frequency
Age
0
50
100
150
200
250
300
350
400
40
60
80
100
120


Income Distribution
0
100000
200000
300000
400000
500000
600000
1